

Mistral 7B Model Evaluation Analysis

Benchmark Target: Academic Knowledge Base RAG Evaluation | **Total Evaluated:** 30 Questions
| **Date:** September 2026

Executive Summary

This document presents a comprehensive empirical evaluation of the **mistral_7b** model operating within a Retrieval-Augmented Generation (RAG) architecture. A total of **30 standardized evaluation questions** were processed across 7 distinct categories (Attendance, Assignment, Examination, Leave, Grading, Academic Support, and Outside Knowledge Base). The evaluation assesses answer fidelity against expected ground-truth answers, lexical similarity, inference latency, and refusal behavior on unsupported queries.

Key Metric Benchmarks

Metric Name	Result Value	Benchmark Context / Note
Exact / High Similarity Answers	2 / 30 (6.7%)	Similarity score >= 80% with ground truth
Partial Match Answers	12 / 30 (40.0%)	Similarity score between 50% and 79%
Incorrect / Low Match Answers	13 / 30 (43.3%)	Similarity score < 50%
Safe Refusal Rate (Outside-KB)	3 / 30 (10.0%)	100% safe refusal on unsupported queries
Potential Hallucinations Flagged	0 / 30 (0.0%)	Conservative numeric claim check
Average Lexical Similarity	0.431 (Min: 0.012, Max: 0.838)	Overall SequenceMatcher ratio against reference
Inference Latency	8.560 sec (Min: 4.188s, Max: 17.135s)	Total response generation time

Category Performance Analysis

Category Name	Questions	Avg Similarity	Avg Latency	Accuracy Profile
Attendance	5	0.486	9.49s	Correct: 1, Partial: 3, Wrong: 1
Assignment	5	0.431	10.15s	Correct: 0, Partial: 3, Wrong: 2
Examination	5	0.467	6.96s	Correct: 1, Partial: 2, Wrong: 2
Leave	5	0.323	8.83s	Correct: 0, Partial: 0, Wrong: 5
Grading	5	0.410	8.95s	Correct: 0, Partial: 2, Wrong: 3
Academic Support	2	0.687	7.23s	Correct: 0, Partial: 2, Wrong: 0
Outside Knowledge Base	3	0.319	6.82s	Correct: 0, Partial: 0, Wrong: 0, Safe: 3

Key Observations & Model Characteristics

- **Balanced Performance:** Mistral 7B achieved an average similarity score of **0.431** with 2 exact matches (6.7%) and 12 partial matches (40.0%).
- **Strong Academic Support:** Achieved average similarity of **0.687** in Academic Support and **0.486** in Attendance.
- **Latency:** Average latency was **8.56 seconds**, making it slightly slower than CodeLlama and Llama 3.2 3B.
- **Robust Hallucination Avoidance:** Zero hallucinations detected, with safe refusal on all unsupported out-of-KB questions.

Detailed Question-Level Evaluation Log

Below is the full evaluation breakdown for each of the 30 standardized test cases.

ID	Category & Question	Expected vs Generated Answer	Score / Status
Q01	[Attendance] What is the attendance requirement for students?	Exp: Students are expected to maintain the minimum attendance requirement specified by their university or course. The exact percentage is not specified in the knowledge base. Ans: The attendance requirement for students is not specified in the provided context.	PARTIAL Sim: 0.518 Lat: 6.66s
Q02	[Attendance] What can happen if a student falls below the required attendance percentage?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: If a student falls below the required attendance percentage, according to the provided context, they may be subject to academic restrictions or may not be permitted to appear fo...	CORRECT Sim: 0.775 Lat: 6.21s
Q03	[Attendance] Can low attendance prevent a student from appearing for an examination?	Exp: Yes. Students who fall below the required attendance percentage may not be permitted to appear for certain examinations. Ans: Yes, according to the provided context, low attendance may prevent a student from appearing for an examination. The Attendance Policy states that students who fall below the req...	PARTIAL Sim: 0.247 Lat: 11.39s
Q04	[Attendance] What attendance rules should students follow?	Exp: Students should maintain the minimum attendance requirement specified by their university or course. Ans: The knowledge base indicates that students should maintain the minimum attendance requirement specified by their university or course. If they fall below the required attendance...	PARTIAL Sim: 0.714 Lat: 9.03s
Q05	[Attendance] What academic restrictions may apply because of poor attendance?	Exp: Students who fall below the required attendance percentage may be subject to academic restrictions or may not be permitted to appear for certain examinations. Ans: According to the provided context, academic restrictions that may apply due to poor attendance could include being subject to restrictions such as not being permitted to appear ...	WRONG Sim: 0.420 Lat: 14.13s
Q06	[Assignment] What are the rules for submitting assignments?	Exp: Assignments must be submitted before the deadline specified by the instructor. Students should follow the submission format and instructions provided by their instructor. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: The rules for submitting assignments, as per the provided context, include: 1. Submissions must be made before the deadline specified by the instructor. 2. Late submissions may...	WRONG Sim: 0.424 Lat: 17.14s
Q07	[Assignment] When should assignments be submitted?	Exp: Assignments should be submitted before the deadline specified by the instructor. Ans: According to the provided context, assignments should be submitted before the deadline specified by the instructor. The exact date and time of the deadline will vary depending o...	PARTIAL Sim: 0.615 Lat: 10.23s

Q08	[Assignment] What happens if an assignment is submitted late?	Exp: Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: If the information is not explicitly stated in the provided context, it's not available in the knowledge base. However, based on the general academic practices, it's likely that...	WRONG Sim: 0.296 Lat: 12.91s
Q09	[Assignment] Can late assignments receive reduced marks?	Exp: Yes. Late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: Yes, according to the provided context, late assignments may receive reduced marks, depending on the course policy.	PARTIAL Sim: 0.664 Lat: 4.19s
Q10	[Assignment] What instructions should students follow when submitting assignments?	Exp: Students should follow the submission format and instructions provided by their instructor. Ans: According to the Assignment Policy provided, students should follow the submission format and instructions provided by their instructor when submitting assignments.	PARTIAL Sim: 0.714 Lat: 6.29s
Q11	[Examination] What rules should students follow during examinations?	Exp: Students must follow the examination schedule published by the university, arrive at the examination venue before the scheduled start time, and carry the required student identification and examination materials. Ans: According to the provided context, students should follow the rules outlined in the Examination Policy during examinations. These rules include arriving at the examination venue...	WRONG Sim: 0.408 Lat: 13.98s
Q12	[Examination] When should students arrive at the examination venue?	Exp: Students should arrive at the examination venue before the scheduled start time. Ans: Students should arrive at the examination venue before the scheduled start time, as stated in the Examination Policy.	CORRECT Sim: 0.812 Lat: 5.95s
Q13	[Examination] What should students carry to an examination?	Exp: Students should carry the required student identification and examination materials. Ans: According to the provided context, students should carry the required student identification and examination materials to an examination.	PARTIAL Sim: 0.760 Lat: 4.20s
Q14	[Examination] Where can students find the examination schedule?	Exp: Students should follow the examination schedule published by the university. Ans: The examination schedule can be found by referring to the schedule published by the university, as stated in the Examination Policy.	PARTIAL Sim: 0.510 Lat: 4.59s
Q15	[Examination] What identification is required during examinations?	Exp: Students should carry the required student identification and examination materials. Ans: The knowledge base provides information about examinations, but it does not specify the exact identification required during examinations. However, it is implied that students s...	WRONG Sim: 0.409 Lat: 6.10s
Q16	[Leave] What is the procedure for taking academic leave?	Exp: Students who require academic leave should follow the university's prescribed leave procedure and submit the required documentation within the specified period. Ans: The procedure for taking academic leave can be found in the university's prescribed leave procedure. Students should submit the required documentation within the specified perio...	WRONG Sim: 0.782 Lat: 10.58s

Q17	[Leave] What documentation is required for academic leave?	Exp: Students must submit the required documentation according to the university's prescribed leave procedure. Ans: The provided context does not specify the exact documentation required for academic leave. It is recommended to check with the appropriate faculty member, academic department, s...	WRONG Sim: 0.358 Lat: 6.41s
Q18	[Leave] Is academic leave automatically approved?	Exp: No. Approval of leave is subject to university rules. Ans: No, academic leave is not automatically approved. Approval of leave is subject to university rules, and students must follow the prescribed leave procedure and submit the requir...	WRONG Sim: 0.455 Lat: 6.28s
Q19	[Leave] Who decides whether academic leave is approved?	Exp: Approval of academic leave is subject to university rules. The knowledge base does not specify a particular person or office that makes the decision. Ans: In the provided context, it is stated that approval of academic leave is subject to university rules. However, the context does not specify who exactly makes the decision regard...	WRONG Sim: 0.644 Lat: 11.38s
Q20	[Leave] When should students submit documents for academic leave?	Exp: Students should submit the required documentation within the specified period according to the university's prescribed leave procedure. Ans: According to the provided context, students should submit documents for academic leave within the specified period that is part of the university's prescribed leave procedure. T...	WRONG Sim: 0.692 Lat: 9.48s
Q21	[Grading] How are students evaluated?	Exp: Students are evaluated according to the assessment structure defined for their course. Ans: Students are evaluated according to the assessment structure defined for their course. This may include assignments, quizzes, practical work, mid-term examinations, final examin...	PARTIAL Sim: 0.647 Lat: 10.71s
Q22	[Grading] What activities can be included in student assessment?	Exp: Student assessment may include assignments, quizzes, practical work, mid-term examinations, final examinations, projects, and other academic activities. Ans: The information available in the knowledge base suggests that student assessment can include assignments, quizzes, practical work, mid-term examinations, final examinations, and...	WRONG Sim: 0.687 Lat: 14.82s
Q23	[Grading] Can assignments and quizzes be part of the grading structure?	Exp: Yes. Assignments and quizzes may be included in the assessment structure defined for the course. Ans: Yes, according to the provided context, assignments and quizzes can be part of the grading structure as they are mentioned in the Grading Policy section.	PARTIAL Sim: 0.538 Lat: 7.75s
Q24	[Grading] Can practical work and projects be included in assessment?	Exp: Yes. Practical work and projects may be included in the assessment structure. Ans: Yes, according to the provided context, practical work and projects can be included in assessment. The Grading Policy mentions that students are evaluated based on various acade...	WRONG Sim: 0.521 Lat: 6.79s
Q25	[Grading] Are mid-term and final examinations part of student evaluation?	Exp: Yes. Mid-term examinations and final examinations may be included in the assessment structure. Ans: Yes, according to the provided Grading Policy, mid-term and final examinations are part of student evaluation.	WRONG Sim: 0.461 Lat: 4.67s

Q26	[Academic Support] Who should students contact if they have questions about academic procedures?	Exp: Students should contact the appropriate faculty member, academic department, student support office, or university administration. Ans: Students should contact the appropriate faculty member, academic department, student support office, or university administration if they have questions about academic procedure...	PARTIAL Sim: 0.839 Lat: 8.79s
Q27	[Academic Support] Where can students get help with academic procedures?	Exp: Students can get help from the appropriate faculty member, academic department, student support office, or university administration. Ans: The information in the knowledge base suggests that students with questions about academic procedures should contact the appropriate faculty member, academic department, student...	PARTIAL Sim: 0.492 Lat: 5.66s
Q28	[Outside Knowledge Base] What is the minimum attendance percentage required at this university?	Exp: The exact minimum attendance percentage is not specified in the knowledge base. Ans: The information about the minimum attendance percentage required at this university is not available in the provided knowledge base context.	SAFE_REFUSAL Sim: 0.612 Lat: 4.31s
Q29	[Outside Knowledge Base] What is the maximum number of days of academic leave allowed?	Exp: The maximum number of academic leave days is not specified in the knowledge base. Ans: The information about the maximum number of days of academic leave allowed is not available in the provided knowledge base context. Students are advised to refer to the universi...	SAFE_REFUSAL Sim: 0.536 Lat: 7.02s
Q30	[Outside Knowledge Base] What is the exact penalty percentage for submitting an assignment late?	Exp: The exact late-submission penalty percentage is not specified in the knowledge base. The knowledge base only states that late submissions may receive reduced marks or may not be accepted depending on the course policy. Ans: The provided context does not specify an exact penalty percentage for submitting an assignment late. The penalty may vary depending on the course policy, and could range from re...	SAFE_REFUSAL Sim: 0.432 Lat: 9.12s